

Yi-Chun Liao

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EDUCATION

Duke University

Ph.D. in Electrical and Computer Engineering

- Advisor: *Prof. Yiran Chen*

Durham, NC

Aug. 2026 – Present

National Taiwan University (NTU)

B.Sc. in Computer Science and Information Engineering (CSIE)

- Overall GPA: 3.90/4.30, Last 60 GPA: 4.09/4.30

Taipei, Taiwan

Sep. 2021 – Jan. 2026

PUBLICATIONS

Yi-Chun Liao, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. “RETENTION: Resource-Efficient Tree-Based Ensemble Model Acceleration with Content-Addressable Memory,” in *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, doi: [10.1109/TCAD.2026.3691288](https://doi.org/10.1109/TCAD.2026.3691288).

Yi-Chun Liao, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. “Practicalizing Tree-Based Model Acceleration with CAM through Model Pruning and Data Placement Optimization.” *ACM/IEEE International Conference on Hardware/Software Codesign and System Synthesis (CODES+ISSS '25)*, September 28–October 3, 2025, Taipei, Taiwan. ACM, New York, NY, USA, 2 pages. doi:[10.1145/3742873.3758342](https://doi.org/10.1145/3742873.3758342).

Hong-Wei Li, Ping-Ting Liu, Bo-Wei Lin, **Yi-Chun Liao**, and Yennun Huang. “IPMES: A Tool for Incremental TTP Detection Over the System Audit Event Stream,” *2024 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, Brisbane, Australia, 2024, pp. 265–273, doi:[10.1109/DSN58291.2024.00036](https://doi.org/10.1109/DSN58291.2024.00036).

RESEARCH EXPERIENCE

Research Intern

University of Notre Dame, Advisor: Prof. X. Sharon Hu

Indiana, United States

May 2025 – Present

- Conduct independent research on realizing scalable in-memory retrieval for retrieval-augmented generation (RAG), co-designing algorithm and hardware to prevent the massive data movement of retrieval phase.
- Conduct independent research on alleviating the capacity constraint of in-memory hyperdimensional computing, adapting the model for hardware constraints and introducing an algorithm for scalable CAM implementations.
- Contribute to the extension of the open-source CAM simulator [CAMASim](#), enabling data mapping optimizations with SPC and ODR, algorithms proposed in RETENTION.

Research Assistant

Academia Sinica, Advisors: Prof. Yuan-Hao Chang and Prof. Tei-Wei Kuo

Taipei, Taiwan

Apr. 2024 – Jun. 2026

- Conducted independent research on minimizing resource consumption for tree-based ensemble model inference, introducing an end-to-end framework [RETENTION](#) to minimize the capacity requirement for CAM-based acceleration.
- Explored and quantified the relationship between capacity redundancy and processing overhead, enabling flexible tradeoffs between controllable accuracy loss, energy consumption, and capacity requirement.
- Reduced capacity requirement by $4.35\times$ to $207.12\times$ with less than 3% accuracy loss and no degradation on throughput and energy efficiency. The first-authored paper was published in TCAD.
- Collaborated on minimizing retrieved tokens for RAG acceleration, exploring the tradeoffs between accuracy, latency, and capacity requirement.

Research Assistant

Academia Sinica, Advisor: Prof. Yennun Huang

Taipei, Taiwan

Jul. 2023 – Jun. 2024

- Collaborated on developing a data analytics tool, [IPMES](#), to detect structural anomalies and trace event patterns within system log streams. The co-authored paper was published in DSN 2024.
- Constructed an automated framework to simulate diverse system workloads and log operating system behavior inside virtual machines. The framework was subsequently used for dataset synthesis.

TEACHING EXPERIENCE

Teaching Assistant

Computer Architecture, Lecturer: *Prof. Chia-Lin Yang*

- Supported the instruction of the junior compulsory course Computer Architecture at NTU, enrolled by over 100 students.
- Led weekly discussion sessions, designed and/or graded assignments, quizzes, and exams.
- Provided voluntary office hours, exceeding 10 hours per week during exam periods.

Taipei, Taiwan

Sep. 2024 – Jan. 2025

Teaching Assistant

APCS Camp

- Provided TA hours for over 300 C++ beginners during two-week crash courses.
- Adapted teaching strategies including programming problem solving and customized course reviews to meet the diverse needs and skill levels of students.

Taipei, Taiwan

Jul. 2022/23 – Aug. 2022/23

HONORS

First Place of 2025 Bachelor Thesis Award

National Taiwan University

- Distinguished through a three-stage selection process and recognized as one of the top 2 in the CSIE Department, the best in the EECS College, and one of the top 6 university-wide.

Taipei, Taiwan

Jun. 2025

Appier Enterprise Award

National Taiwan University

- Received the Most Potential Award from Appier in the departmental undergraduate research exhibition.

Taipei, Taiwan

Jun. 2025

Dean's List Award

National Taiwan University

- Ranked 1/155 in the department for both Spring and Fall 2024 semesters.

Taipei, Taiwan

Mar. 2025

Irving T. Ho Memorial Scholarship

National Taiwan University

- Awarded to 3 students in the department for academic and research excellence.

Taipei, Taiwan

Feb. 2025

First Place of AUO Enterprise Award

MakeNTU 2024 (Nationalwide Makerthon)

- Designed an interactive in-car RPG game for AUO smart windows to enhance travel experience, which was recognized as the best work for AUO Enterprise Award among 43 teams.

Taipei, Taiwan

May 2024

EXTRACURRICULAR ACTIVITIES

President

NTU CSIE Department Student Association

- Directed the student council of the CSIE department with over 50 members.
- Promoted student welfare and represented over 600 students in faculty meetings, contributing to departmental decisions.

Taipei, Taiwan

Aug. 2023 – Jul. 2024

Vice President

NTU EECS College Student Association

- Represented over 1500 students in college faculty meetings and helped shape student-related policies.

Taipei, Taiwan

Aug. 2023 – Jul. 2024

President

NTU Cancer Children Service Club

- Led a 50-member service club, managing budget allocation and arranging educational and recreational activities.
- Worked with Childhood Cancer Foundation of R.O.C., organizing voluntary services to reduce the burden of parents.

Taipei, Taiwan

Feb. 2023 – Jul. 2023

Vice Coordinator

NTU CSIE Camp

- Organized a 6-day camp for over 100 high school students with a 50-member staff team.
- Designed activities to introduce students to the CSIE department and provide insights into university life.

Taipei, Taiwan

Feb. 2023 – Jul. 2023